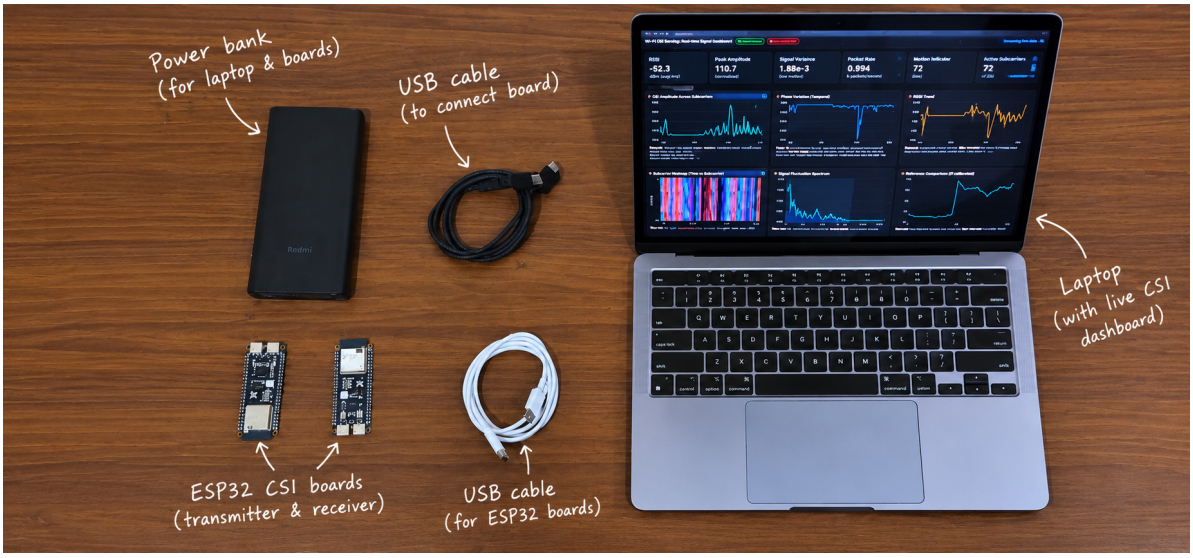
VIT-AP University

School of Computer Science and Engineering (SCOPE)

ESP32-Based Wi-Fi Channel State Information for Human Presence Sensing

IoT Subject Project Report



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Declaration

We declare that this report presents the design, implementation, and preliminary examination of our ESP32-based Wi-Fi Channel State Information (CSI) sensing prototype for the Internet of Things (IoT) subject. The report describes the available hardware setup, recorded data, visualization interface, and limitations of the current experimental evidence.

The results and interpretations are presented according to the available project materials. Proposed validation steps are identified as future work rather than represented as completed experiments.

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(Digitally signed)
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Abstract

Wi-Fi Channel State Information (CSI) can provide measurements of wireless channel behaviour that may change when objects or people alter the propagation environment. This project explores an ESP32-based setup for collecting CSI records and inspecting their signal characteristics through a software dashboard. The intended application is exploratory human-presence sensing without relying on a camera or a wearable device.

The available log was examined to determine its record structure and basic measurement statistics. It contains 16,660 records marked as `CSI_DATA`, each with 25 comma-separated fields. The examined channel-related field has the value 11 throughout the log, the reported CSI-length field is 256, and the RSSI values range from -73 dBm to -29 dBm, with a mean of approximately -42.45 dBm. The dashboard provides visualizations for signal measurements and signal-derived indicators.

These observations demonstrate that a substantial set of records is available for exploratory inspection. They do not, by themselves, establish a validated human-presence detector. The available log does not contain synchronized presence/absence labels, and the supplied dashboard screenshot indicates that calibration is required and no data are arriving from the receiver in that screenshot. Accordingly, classification accuracy and other detection-performance metrics are not claimed. The report documents the prototype, its preliminary data characteristics, limitations, and the controlled evaluation needed for future development.

Keywords: Internet of Things, ESP32, Wi-Fi sensing, Channel State Information, RSSI, human presence sensing, signal analysis.

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Abbreviations

CSI	Channel State Information
IoT	Internet of Things
RSSI	Received Signal Strength Indicator
Wi-Fi	Wireless networking technology based on IEEE 802.11
OFDM	Orthogonal Frequency-Division Multiplexing
ESP32	Espressif microcontroller family used in the prototype
CSV	Comma-Separated Values
FPR	False-Positive Rate
TPR	True-Positive Rate
F1-score	Harmonic mean of precision and recall

Chapter 1

Introduction

1.1 Background

The Internet of Things connects physical devices, sensors, and software services to collect information and support monitoring or automation. Many IoT applications depend on detecting changes in the surrounding environment, including occupancy, movement, temperature, light, and other contextual conditions.

Human-presence sensing is useful in smart buildings, energy management, room-occupancy monitoring, and context-aware systems. Conventional solutions include cameras, passive infrared sensors, and wearable devices. Their practical limitations vary: cameras may raise privacy concerns, passive infrared sensors may be less responsive to some stationary situations, and wearables require the user to carry a device.

Wi-Fi sensing offers another research direction. Wireless signals interact with walls, furniture, people, and other objects as they travel through an environment. Changes in propagation paths can affect the measured wireless channel. Channel State Information provides a more detailed view of the channel than a single received signal-strength measurement and has therefore been studied for sensing applications [1, 2].

1.2 Problem Statement

The project investigates whether measurements collected with an ESP32-based CSI setup can be recorded and visualized in a form that supports exploration of human-presence-related signal changes. The immediate engineering problem is to establish a repeatable acquisition and inspection workflow and determine what additional experiments are needed before a reliable detector can be claimed.

A change in RSSI or CSI is not uniquely caused by human presence. Device position, interference, multipath propagation, and other environmental conditions can also affect measurements. Signal visualization is therefore a useful exploratory step, but it is not

sufficient evidence of validated presence detection.

1.3 Objectives

1. Describe the ESP32-based hardware and software workflow.
2. Inspect the structure of the available CSI log.
3. Summarize basic RSSI and CSI-related field observations.
4. Explain the role of the dashboard in signal inspection.
5. Identify the limitations of the current evidence.
6. Define a controlled testing procedure for future validation.

1.4 Scope

This report covers the prototype setup, data logging, exploratory signal analysis, dashboard visualizations, potential applications, and a proposed evaluation plan. It does not claim a completed machine-learning classifier, a fully validated real-time detector, or demonstrated accuracy through walls.

Chapter 2

Technical Background

2.1 Wi-Fi Channel State Information

CSI describes aspects of the wireless channel response. In a multipath environment, a transmitted signal may reach the receiver through several paths. Reflection, scattering, and obstruction can change the relative contributions of those paths.

For a subcarrier k, a simplified complex channel measurement is

$$H_k(t) = |H_k(t)|e^{j\phi_k(t)},$$

where $|H_k(t)|$ represents amplitude and $\phi_k(t)$ represents phase at time t. This is conceptual background; it does not imply that every component has been calibrated or analysed in this prototype.

CSI-based sensing has been investigated for activities and environmental changes, but measurement quality and generalization remain important challenges [1, 2].

2.2 Received Signal Strength Indicator

RSSI is a received-signal-strength measurement reported by the wireless system. It is useful for monitoring coarse changes in received power, but it does not describe all of the frequency-dependent structure available in CSI.

RSSI may change due to movement, distance, orientation, interference, or environmental conditions. It should be interpreted as one signal indicator rather than a direct measurement of human presence.

2.3 Amplitude and Phase Features

Amplitude variation can be examined across records and subcarriers. Temporal statistics, differences between adjacent measurements, and correlations between records may also help characterize changes in the channel.

Phase can potentially contribute additional information. Its reliability depends on acquisition characteristics, hardware behaviour, and appropriate processing. Research has explored combining amplitude and phase for activity recognition, but such methods require their own implementation and evaluation [3].

2.4 From Signal Changes to Presence Detection

A presence detector requires a defined target condition and a repeatable evaluation process. At minimum, data should be collected under labelled presence and absence conditions. A threshold or classifier should then be assessed on data not used to choose its parameters. The current unlabelled log cannot establish performance metrics such as accuracy, precision, recall, or F1-score.

Chapter 3

System Architecture and Hardware

3.1 Hardware Components

The setup documented for this project contains two ESP32-based development boards, USB cables, a power bank, and a laptop displaying the CSI dashboard. The exact board variants and their individual transmitter/receiver roles should be confirmed against the board labels and firmware configuration before being stated as precise hardware specifications.

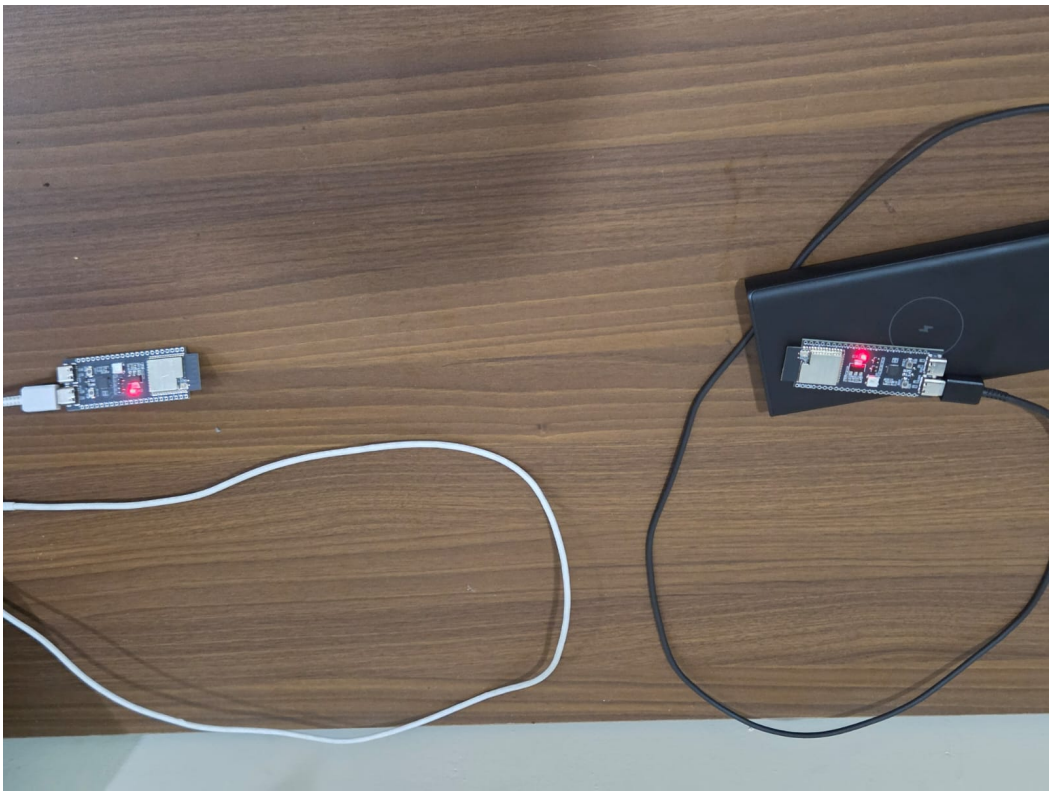


Figure 3.1: ESP32 hardware used in the prototype.



Figure 3.2: Prototype setup showing the hardware arrangement.

3.2 Conceptual Workflow

The intended workflow comprises wireless measurement, CSI acquisition, record logging, data inspection, and dashboard visualization. These stages describe the system concept and should not be interpreted as proof that every stage was continuously operating at the time of the supplied dashboard screenshot.

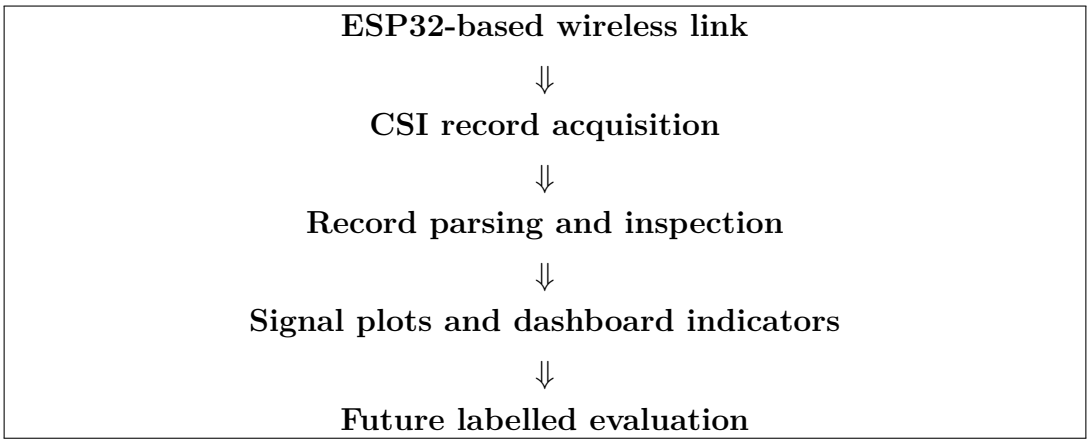


Figure 3.3: Conceptual workflow of the CSI sensing prototype.

3.3 Acquisition and Logging

The available log uses the marker `CSI_DATA`. Each parsed record contains 25 comma-separated fields. The inspected records consistently report the value 11 in the examined channel-related field and 256 in the reported CSI-length field.

The final field contains a quoted array of integer values, with 256 entries in each inspected record. The length of this array should not automatically be equated with the

number of active physical OFDM subcarriers. That interpretation requires the specific firmware format and device configuration.

3.4 Dashboard Layer

The dashboard is intended to help inspect measurements and signal-derived indicators through charts and summary panels. A metric labelled as a motion indicator, for example, should not be treated as a validated motion classifier without documented feature definitions, thresholds, and testing.

Chapter 4

Implementation and Data Processing

4.1 Data Inspection Procedure

The available log was inspected by parsing records with CSV-aware handling so that commas inside quoted array fields would not be mistaken for separators between fields. The inspection checked the record marker, number of fields, selected channel-related values, reported CSI length, RSSI values, and the structure of the integer array in the final field.

This is a structural and descriptive inspection. It does not establish that every record represents a different human activity, a unique environment, or a labelled presence event.

4.2 Observed Log Characteristics

Table 4.1 summarizes the values verified from the available log.

Table 4.1: Descriptive summary of the inspected CSI log.

Property	**Observed value**
Number of records	16,660
Record marker	`CSI_DATA`
Fields per record	25
RSSI minimum	−73 dBm
RSSI maximum	−29 dBm
Mean RSSI	Approximately −42.45 dBm
Examined channel-related field	11 throughout the inspected log
Reported CSI-length field	256 throughout the inspected log
Integer entries in final field	256 per inspected record
Array parsing errors	0 in the inspected records

4.3 Potential Processing Steps

A future repeatable processing pipeline could include:

1. **Record validation:** verify field count, marker, timestamps if available, and array parsing.
2. **Signal extraction:** decode CSI values using the correct firmware format and document their interpretation.
3. **Preprocessing:** identify missing records, remove invalid measurements, and apply only justified filtering.
4. **Feature calculation:** calculate defined statistics such as amplitude variation or temporal differences.
5. **Calibration:** establish a baseline under a documented environment and device configuration.
6. **Evaluation:** compare predictions with independent, time-aligned ground-truth labels.

These steps are recommended future work, not a claim that all operations have already been implemented.

4.4 Reproducibility Considerations

A repeatable experiment should record the board model, firmware version, wireless channel, transmitter/receiver placement, sampling configuration, room layout, and environmental conditions. Presence or absence should be labelled with timestamps so labels can be aligned with signal records.

Chapter 5

Results and Dashboard Analysis

5.1 Descriptive Results

The structural inspection confirms that the supplied log contains 16,660 parseable records with a consistent 25-field layout. The observed RSSI values span 44 dB, from −73 dBm to −29 dBm, and have a mean of approximately −42.45 dBm.

The examined channel-related field remained at 11, and the reported CSI-length field remained at 256 throughout the inspected records. The final field contained 256 integer entries per record. These observations characterize the log format and selected measurements; they are not detection-performance results.

5.2 Dashboard Screenshot

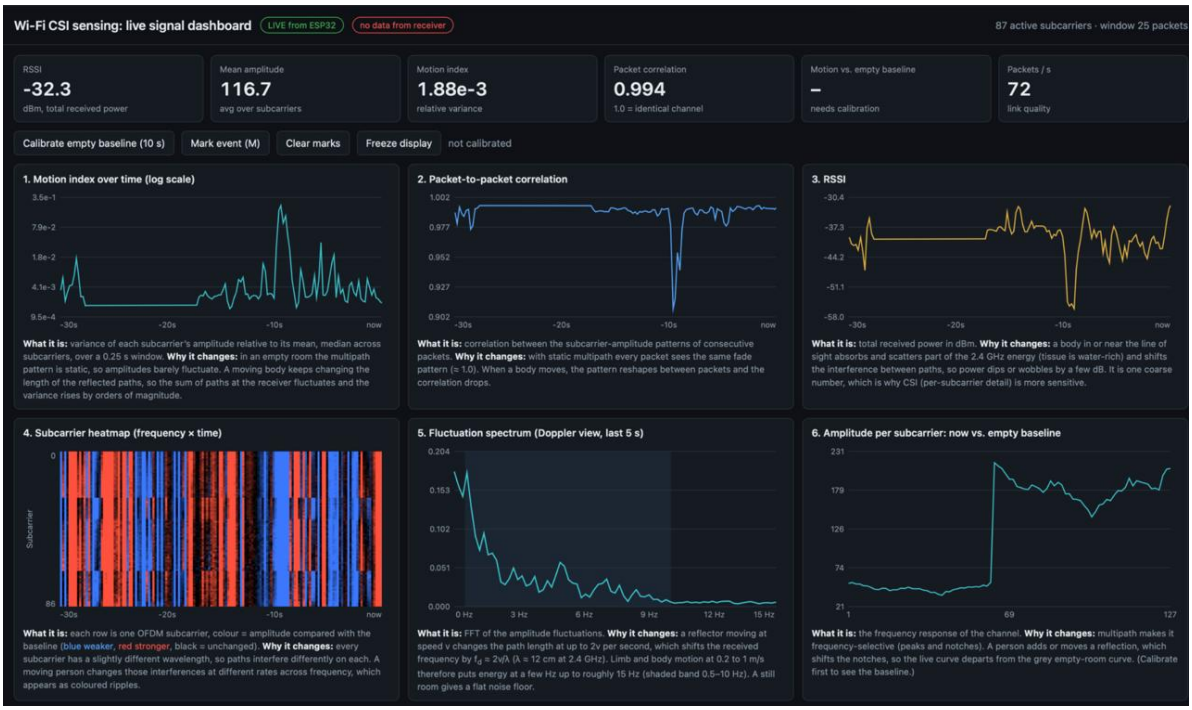


Figure 5.1: Dashboard interface used for exploratory inspection of Wi-Fi CSI-related measurements.

The supplied screenshot includes summary values and visualizations such as RSSI-related information, amplitude-related plots, temporal variations, a heatmap, and other signal-derived indicators. The screenshot indicates that calibration is required and that no data are arriving from the receiver in that view. It should therefore be treated as evidence of the interface, not proof of continuous live acquisition or validated presence detection.

5.3 Interpretation of the Visualizations

Time-series plots can reveal visible fluctuations, while a heatmap can help show differences across a sequence of measurements. The meaning of each indicator depends on how it is calculated. A plotted motion index, packet-correlation value, or subcarrier-related metric is not automatically equivalent to a presence probability.

5.4 What the Current Evidence Supports

The current evidence supports these statements:

- A structured log containing 16,660 CSI-marked records is available.

- The inspected records have a consistent 25-field layout.
- RSSI values and selected CSI-related fields can be summarized.
- A dashboard interface has been prepared to visualize signal-related measurements.

The current evidence does not establish classification accuracy, precision, recall, F1-score, false-positive rate, reliable detection of stationary people, or generalization to different rooms and wall materials. Those require labelled experiments and independent evaluation.

Chapter 6

Testing and Validation Plan

6.1 Need for Controlled Testing

A human-presence sensing prototype should be evaluated under repeatable conditions. Visual changes in a chart may motivate a hypothesis, but they are not enough to show that the system distinguishes presence from absence reliably.

The current log has no synchronized ground-truth labels identifying when a person is present or absent. Detector performance metrics cannot therefore be calculated from this log alone.

6.2 Recommended Experimental Conditions

A future collection plan should include repeated trials for:

- An empty environment with no intentional human movement.
- A person entering and leaving the sensing area.
- A person remaining stationary for a defined interval.
- A person moving within the sensing area.
- Repeated trials with the same hardware positions.
- Additional trials after changing the room or device placement.

These are recommendations for future testing, not a claim that all conditions have already been measured.

6.3 Ground-Truth Labelling

Each trial should have a documented start time, end time, and presence state. The label should be aligned with the recorded CSI data. A trial log should also note the room, device placement, human activity, and relevant environmental changes.

Training and evaluation records should be separated to avoid reporting performance on the same trials used to choose thresholds or model parameters.

6.4 Suggested Evaluation Metrics

Once labelled test data are available, the following metrics can be calculated:

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP denotes true positives, FP false positives, and FN false negatives. Accuracy and false-positive rate may also be reported, provided the class distribution and evaluation procedure are documented. No numerical values for these metrics are reported because the necessary ground-truth labels and validated predictions are not available in the inspected log.

Chapter 7

Applications, Benefits, and Limitations

7.1 Potential Applications

If developed and validated further, Wi-Fi CSI sensing may be relevant to:

- **Smart buildings:** occupancy-related automation and room-use monitoring.
- **Energy management:** occupancy-related inputs to lighting or HVAC control.
- **Context-aware IoT:** environmental signal features for monitoring systems.
- **Privacy-conscious sensing research:** alternatives to camera-based observation in selected settings.

These are potential application areas; the present prototype has not been validated for deployment in these settings.

7.2 Potential Benefits

CSI-based approaches can use wireless channel measurements rather than requiring a person to wear a dedicated sensor. They also offer a research opportunity to investigate signal changes using accessible development hardware. These potential benefits must be balanced against calibration, evaluation, and environmental sensitivity.

7.3 Current Limitations

1. The available log does not contain ground-truth presence/absence labels.
2. The dashboard screenshot reports a calibration requirement and no incoming receiver data.

3. The precise interpretation of all log fields depends on the firmware format.
4. The reported array length should not be assumed to equal the number of active physical subcarriers.
5. No validated classification metrics are available.
6. Generalization across rooms, wall materials, device placements, and people has not been established.

Chapter 8

Future Enhancements

8.1 Reliable Acquisition

The first priority is to confirm a stable connection between the receiver and dashboard. Acquisition status should be visible, and logs should include timestamps or another reliable means of aligning measurements with experimental events.

8.2 Calibration and Feature Documentation

The dashboard should document the calculation and interpretation of each displayed feature. Calibration should be conducted under a defined baseline condition, with the configuration and procedure recorded so results can be reproduced.

8.3 Labelled Dataset Collection

A controlled dataset should include repeated presence and absence trials, stationary and moving conditions, and documented environmental settings. A clear labelling protocol will make it possible to test whether candidate features distinguish the conditions of interest.

8.4 Model Development and Evaluation

After sufficient labelled data have been collected, simple baseline methods can be evaluated before more complex models are considered. Any threshold or classifier should be evaluated on independent trials. Results should report the data split, class distribution, metrics, and limitations.

8.5 Environmental Robustness

Future work should investigate how performance changes with device separation, orientation, room layout, people, and obstructions. Through-wall sensing, if pursued, should be treated as a separate experimental condition and tested directly rather than inferred from ordinary indoor measurements.

Chapter 9

Conclusion

This project documents an ESP32-based Wi-Fi CSI sensing prototype and its dashboard-based measurement inspection workflow. The available log contains 16,660 CSI-marked records with 25 fields per record. The inspected RSSI values range from -73 dBm to -29 dBm, with a mean of approximately -42.45 dBm. The examined channel-related field remains at 11, and the reported CSI-length field is 256 throughout the inspected log.

The hardware setup and dashboard provide a foundation for further exploration of CSI-related signal changes. Nevertheless, the available evidence is descriptive rather than a validation of human-presence detection. The supplied dashboard screenshot indicates that calibration is required and that no data are arriving from the receiver in that view, while the log lacks ground-truth presence/absence labels.

The next steps are to establish reliable acquisition, document feature calculations, collect labelled trials under controlled conditions, and evaluate the system on independent data. This would allow the project to progress from signal visualization toward a defensible assessment of human-presence sensing.

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